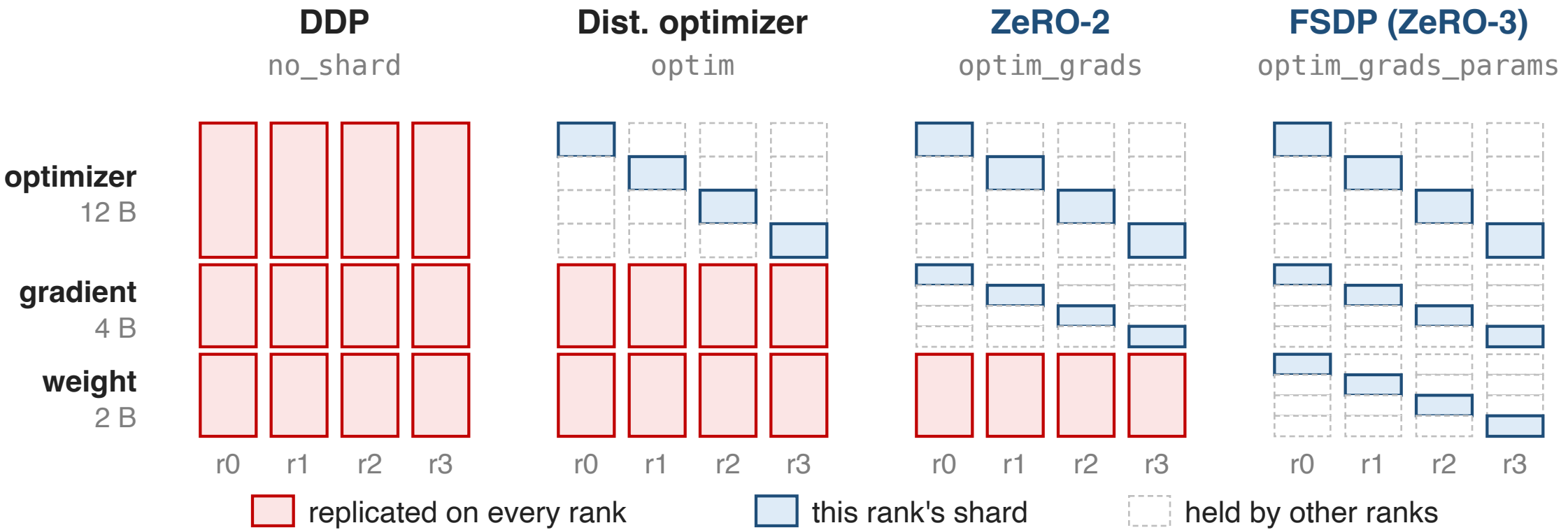
DDP
no_shard
Dist. optimizer
optim
ZeRO-2
optim_grads
FSDP (ZeRO-3)
optim_grads_params
optimizer
12 B
gradient
4 B
weight
2 B
r0
r1
r2
r3
replicated on every rank
this rank's shard
held by other ranks